

# Wireless FM Microphone

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Presented here is a simple wireless microphone with which you can transmit the audio signal to any clear spot in 88-108MHz commercial FM band. The microphone has a range between 15m and 45m depending on the type and location of the receiver's antenna. The unit produces high speech quality with low current drain. A conventional 9V battery powers the unit. The antenna length has been limited to keep radiation within the Federal Communications Commission (FCC) regulation for wireless microphones.

## Circuit and working

The circuit diagram of a wireless FM microphone is shown in Fig. 1. The circuit is built around a speaker (LS1),

audio transformer (X1), op-amp 741 (IC1) and npn transistor 2N3904 (T1). Coil L1 is built with four turns of 18SWG enamelled copper wire. The diameter and length of the coil should be 6.35mm and 25.4mm, respectively. The coil is tapped at the second turn from the ground to connect with transistor T1 as shown in Fig. 1. Two readymade inductors, L2 and L3, are available for the values mentioned in Fig. 1. The antenna is built with a 63.5mm long brass rod that has a diameter of 4.7mm.

## Test Points

| Test point | Details             |
|------------|---------------------|
| TP0        | GND                 |
| TP1        | 12V                 |
| TP2        | Carrier oscillation |

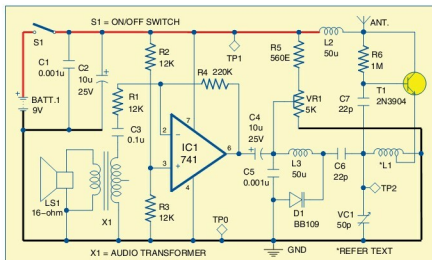


Fig. 1: Circuit diagram of the wireless FM microphone



To use this wireless microphone, tune your FM receiver to a quiet spot and, if possible, turn off the automatic frequency control (AFC) of the receiver to prevent it from tuning automatically to a strong adjacent station. Turn on power switch S1 and speak into the speaker, whilst adjusting trimmer capacitor VC1, until your voice is heard through the FM radio. Adjust potentiometer VR1 for the desired modulation level.

Working of the circuit is simple. Once the transmitter is tuned and you speak into the speaker, your sound

## PARTS LIST

### Semiconductors:

- IC1 - 741 operational amplifier
- T1 - 2N3904 npn transistor
- D1 - BB109 varicap diode

### Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R3 - 12-kilo-ohm
- R4 - 220-kilo-ohm
- R5 - 560-ohm
- R6 - 1-mega-ohm
- VR1 - 5-kilo-ohm preset

### Capacitors:

- C1, C5 - 0.001 $\mu$ F ceramic disk
- C2, C4 - 10 $\mu$ F, 25V electrolytic
- C3 - 0.1 $\mu$ F ceramic disk
- C6, C7 - 22pF ceramic disk
- VC1 - 50pF trimmer capacitor

### Miscellaneous:

- S1 - On/off switch
- L1 - Coil to be made (see text)
- L2, L3 - 50 $\mu$ H inductor
- LS1 - 16-ohm speaker
- X1 - Audio transformer
- BATT1 - 9V battery
- 8-pin IC base
- Antenna



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